MAT 526 — Independent Project

Binary Fission as a Homogeneous Branching Process

A Mathematical Exposition with Simulation

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1 Introduction

In biology, fission is the division of a single entity into two or more separate entities resembling the original. The primary mode of reproduction for bacteria is through binary fission, in which a single organism produces two daughter cells. Suppose the reproduction of each individual cell is independent. How can we track the proliferation of the colony and determine whether it will ever go extinct?

The aim of this paper is to introduce probability generating functions (PGFs) and the homogeneous branching process as a means of modeling the extinction probability of a bacterial colony given certain biological parameters. I derive closed-form expressions for the extinction probability under a Bernoulli $\{0, 2\}$ offspring distribution, and illustrate the results through a simulation.

2 Mathematical Foundations

2.1 The Homogeneous Branching Process

The development in this section follows the treatment of homogeneous branching processes given in Gaudreau Lamarre [2026].

Let C be a random variable taking values in $\mathbb{S} = \{0, 1, 2, \ldots\}$, with probability mass function $\mathbb{P}[C = x]$, $x \in \mathbb{S}$.
We refer to P_C as the offspring distribution.

Definition 2.1. A stochastic process $\{B(t) : t = 0, 1, 2, \ldots\}$ with state space $\mathbb{S}$ is a homogeneous Markov chain if its transition probabilities $p(x, y) = \mathbb{P}[B(t+1) = y \mid B(t) = x]$ do not depend on t.

Definition 2.2. The homogeneous branching process with offspring distribution P_C is the homogeneous Markov chain $B = \{B(t) : t = 0, 1, 2, \ldots\}$ with state space $\mathbb{S}$ and transition probabilities

$$p(x, y) = \begin{cases} 1 & \text{if } x = y = 0, \\ 0 & \text{if } x = 0,\ y \neq 0, \\ \mathbb{P}(C_1 + C_2 + \cdots + C_x = y) & \text{otherwise,} \end{cases} \tag{1}$$

where $C_1, C_2, \ldots$ are independent copies of C and state 0 is absorbing.

Remark. This process is widely known in the literature as the *Galton–Watson branching process*. We refer to it as the homogeneous branching process throughout.

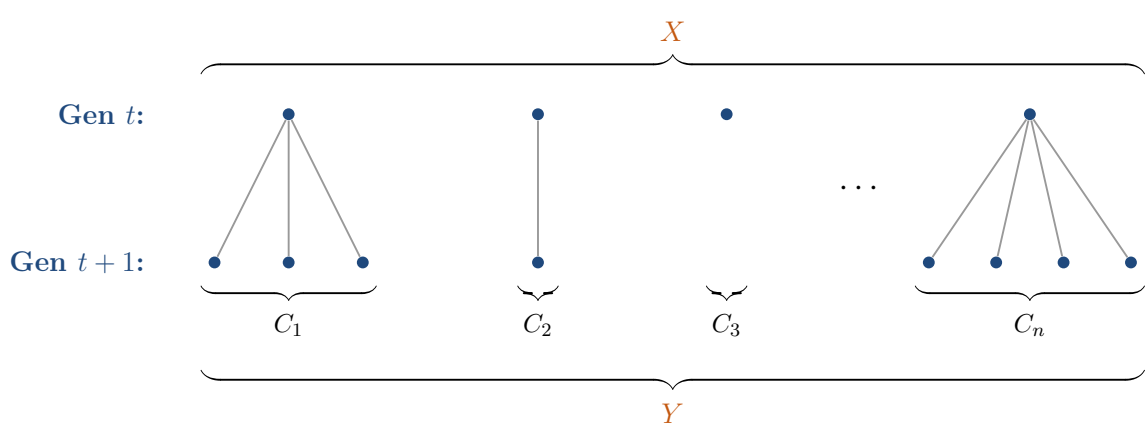

Figure 1: Branching structure of a single generation where each of X individuals independently produces offspring according to P_C

2.2 Probability Generating Functions

The transition structure in (1) tells us what the process is, but to compute extinction probabilities directly from it would require an infinite dimensional transition matrix. The right tool for tracking how probability mass evolves across generations is the probability generating function.

Definition 2.3. The probability generating function (PGF) of P_C is

$$P_C(z) = \mathbb{E}\big[z^C\big] = \sum_{x=0}^{\infty} z^x \, \mathbb{P}[C = x], \qquad z \in [0,1]. \tag{2}$$

Remark. Standard properties: $P_C(1) = 1$, $P_C'(1) = \mu = \mathbb{E}[C]$, and $P_C''(1) = \mathbb{E}[C(C-1)]$.

2.3 The Extinction Probability

Definition 2.4. Starting from x ancestors, the extinction probability is

$$h(x) = \mathbb{P}\left[\bigcup_{t=1}^{\infty}\{B(t) = 0\} \;\middle|\; B(0) = x\right]$$

Remark. A consequence of the independence of the offspring distribution is that extinction probabilities are also independent and identically distributed.
Thus for any $x \geq 2$, $h(x) = h(1)^x$.

Definition 2.5. The mean offspring number is

$$\mu = \mathbb{E}[C] = \sum_{x=0}^{\infty} x\mathbb{P}(C = x) = P_C'(1).$$

Theorem 2.6. The extinction probability $h(1)$ is the smallest fixed point of the PGF $P_C \in [0,1]$:

$$h(1) = \min\{z \in [0,1] : P_C(z) = z\}.$$

Moreover, if $\mathbb{P}(C = 1) < 1$, then $h(1) = 1$ if $\mu \leq 1$, and $h(1) < 1$ if $\mu > 1$.

Remark. In the case every individual produces exactly one offspring, $\mathbb{P}(C = 1) = 1$, the population is constant and never reaches zero, $h(x) = 0$, for all $x \geq 1$

Proof.

Claim 1. $h(1)$ is a solution of $z = P_C(z)$.

By first-step analysis:

$$h(1) = \sum_{x=0}^{\infty} h(x)\, p(1,x) = \sum_{x=0}^{\infty} h(1)^x\, p(1,x) = \sum_{x=0}^{\infty} h(1)^x\, \mathbb{P}(C = x) = P_C(h(1))\,,$$

so $h(1)$ satisfies $z = P_C(z)$.

Claim 2. $z = P_C(z)$ has exactly one or two solutions in $[0,1]$.

Recall $P_C(z) = \sum_{x=0}^{\infty} z^x \mathbb{P}(C = x)$. We examine the shape of P_C on $[0,1]$.

Note the boundary values:

$P_C(0) = \mathbb{P}(C = 0) \in [0,1]$, $P_C(1) = \sum_{x=0}^{\infty} \mathbb{P}(C = x) = 1$.

Differentiating term by term (justified by uniform convergence on $[0,1]$):

$P_C'(z) = \sum_{x=0}^{\infty} x\, z^{x-1} \mathbb{P}(C = x) \geq 0$ for $z \in [0,1]$, so P_C is non-decreasing on $[0,1]$.

Differentiating again: $P_C''(z) = \sum_{x=2}^{\infty} x(x-1)\, z^{x-2} \mathbb{P}(C = x) \geq 0$ for $z \in [0,1]$,

so P_C is convex (concave up) on $[0,1]$.

Claim 3. Suppose that $\mathbb{P}(C = 1) < 1$. If $\mu \leq 1$, then $z = 1$ is the only solution in $[0,1]$.

From above, $P_C'(1) = \sum_{x=1}^{\infty} x\, \mathbb{P}(C = x) = \mathbb{E}[C] = \mu$. The slope of P_C at $z = 1$ equals μ.

- If $\mu \leq 1$: the curve $P_C(z)$ arrives at the point $(1,1)$ with slope at most 1, meaning it approaches from above the diagonal $y = z$ on $(0,1)$. Thus $P_C(z) > z$ for all $z \in [0,1)$, and the only fixed point in $[0,1]$ is $z = 1$.
- If $\mu > 1$: the curve arrives at $(1,1)$ from below the diagonal, so by convexity there must be a second intersection strictly inside $(0,1)$.

Remark. The case $\mu > 1$ requires $P(C = 0) > 0$. In the binary fission model, $P(C = 0) = 1 - p > 0$ for all finite antibiotic concentrations c, so this condition is always satisfied.

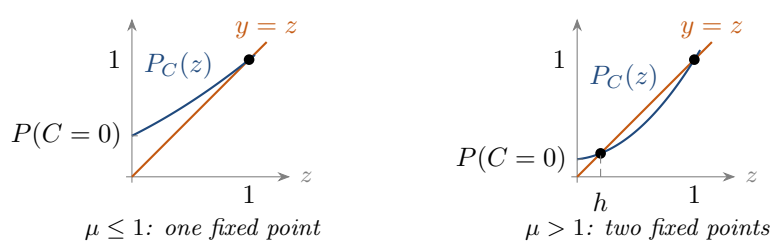

Figure 2: PGF curve versus the diagonal in the subcritical and supercritical cases.

Claim 4. If $\mu > 1$, then $h(1)$ is the *smallest* fixed point of P_C in $[0, 1]$.

Define the finite-horizon extinction probabilities $\mathcal{Z}_n(x) = \mathbb{P}[B(n) = 0 \mid B(0) = x]$, the probability that the process is extinct by generation n starting from x ancestors. Observe that (i) $h(x) = \lim_{n\to\infty} \mathcal{Z}_n(x)$, since extinction by time n increases to eventual extinction, and (ii) $\mathcal{Z}_n(x) = \mathcal{Z}_n(1)^x$ by the same independence argument that gives $h(x) = h(1)^x$.

By first-step analysis and the Markov property:

$$\begin{aligned}\mathcal{Z}_n(1) &= \sum_{x=0}^{\infty} \mathbb{P}[B(n) = 0 \mid B(1) = x,\, B(0) = 1]\, p(1, x) = \sum_{x=0}^{\infty} \mathbb{P}[B(n) = 0 \mid B(1) = x]\, \mathbb{P}(C = x) \\ &= \sum_{x=0}^{\infty} \mathcal{Z}_{n-1}(x)\, \mathbb{P}(C = x) = \sum_{x=0}^{\infty} \mathcal{Z}_{n-1}(1)^x\, \mathbb{P}(C = x) = P_C(\mathcal{Z}_{n-1}(1))\,.\end{aligned}$$

So $\{\mathcal{Z}_n(1)\}$ satisfies $\mathcal{Z}_n(1) = P_C(\mathcal{Z}_{n-1}(1))$ with $\mathcal{Z}_1(1) = \mathbb{P}(C = 0) = P_C(0)$.
Since $P_C(0) \le 1 = P_C(1)$ and P_C is non-decreasing, the sequence is non-decreasing and bounded above by 1, hence convergent. Its limit $\ell = \lim_{n\to\infty} \mathcal{Z}_n(1)$ satisfies $\ell = P_C(\ell)$, so ℓ is a fixed point of P_C. Because the sequence starts at $\mathcal{Z}_1(1) = P_C(0)$, which lies below any fixed point $q^* \ge P_C(0)$, and each iterate remains below q^* (since P_C is non-decreasing), the limit ℓ is the smallest fixed point of P_C in $[0, 1]$, i.e. $\ell = h(1)$. See Figure 3.

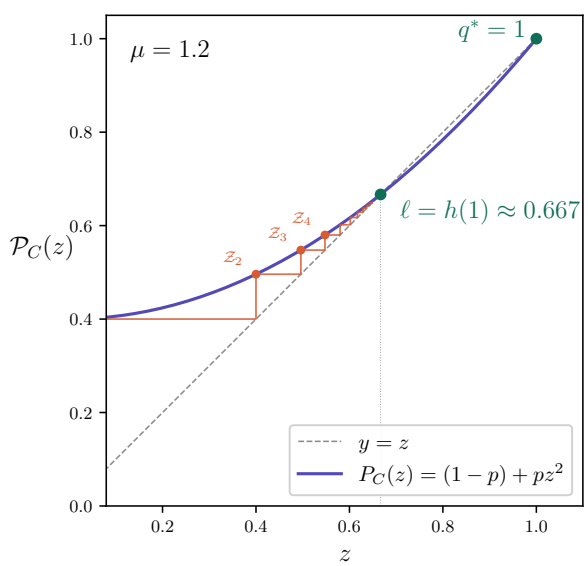

Figure 3: Cobweb iteration of $\mathcal{Z}_n(1) = P_C(\mathcal{Z}_{n-1}(1))$ converging to $\ell = h(1)$, where $p = 0.6$.

Combining all four claims: $h(1)$ is the smallest fixed point of P_C in $[0, 1]$; it equals 1 when $\mu \le 1$, and is strictly less than 1 when $\mu > 1$. $\square$

3 The Binary Fission Model

3.1 Setup

Assumption 3.1. Each bacterium independently divides into two daughters with probability p, or dies with probability $1-p$, where

$$p = \frac{1}{1+\alpha c^{\beta}}, \qquad \alpha, \beta > 0,$$

and $c > 0$ is the antibiotic concentration. The dose–response form and pharmacodynamic parameters are taken from Bogdanov et al. [2022].

3.2 PGF, Mean, and Critical Concentration

From Assumption 3.1 we have

$$\mathbb{P}(C=0) = 1-p, \qquad \mathbb{P}(C=2) = p.$$

Thus the probability generating function is

$$\begin{aligned} P_C(z) &= \sum_{x=0}^{\infty} z^x \mathbb{P}(C=x) \\ &= z^0 \mathbb{P}(C=0) + z^2 \mathbb{P}(C=2) \\ &= (1-p) + pz^2, \end{aligned}$$

and the mean offspring is

$$\mu = \mathbb{E}[C] = 0 \cdot (1-p) + 2p = 2p = \frac{2}{1+\alpha c^{\beta}}.$$

By Theorem 2.6, certain extinction requires $\mu \leq 1$, i.e.

$$\frac{2}{1+\alpha c^{\beta}} \leq 1 \implies 1+\alpha c^{\beta} \geq 2 \implies c^{\beta} \geq \frac{1}{\alpha}.$$

This gives the critical concentration

$$c^* = \left(\frac{1}{\alpha}\right)^{1/\beta}, \tag{3}$$

above which the antibiotic guarantees eventual extinction of the colony.

3.3 Extinction Probability

For $c < c^*$ (i.e. $\mu > 1$), by Theorem 2.6 the extinction probability $h(1)$ is the smallest fixed point of $P_C(z)$ in $[0, 1]$. We solve $z = P_C(z)$:

$$z = (1 - p) + pz^2.$$

Substituting in the value of p:

$$z = \left(\frac{\alpha c^\beta}{1 + \alpha c^\beta}\right) + z^2 \left(\frac{1}{1 + \alpha c^\beta}\right).$$

Multiplying through by $(1 + \alpha c^\beta)$:

$$z(1 + \alpha c^\beta) = \alpha c^\beta + z^2$$

Rearranging:

$$z^2 - z(1 + \alpha c^\beta) + \alpha c^\beta = 0.$$

This factors cleanly as

$$(z - 1)(z - \alpha c^\beta) = 0,$$

yielding two roots:

$$z_1 = 1, \qquad z_2 = \alpha c^\beta.$$

Remark. $z = 1$ is always a fixed point of any PGF since $P_C(1) = 1$ by definition. The interesting root is z_2.

Since $c < c^*$ implies $\alpha c^\beta < 1$, we have $z_2 < z_1$, so the smallest fixed point in $[0, 1]$ is $z_2 = \alpha c^\beta$. Therefore

$$h(1) = \alpha c^\beta, \tag{4}$$

and for a colony of x initial bacteria,

$$h(x) = h(1)^x = \left(\alpha c^\beta\right)^x, \quad c < c^*. \tag{5}$$

3.4 Biological Interpretation & Clinical Application

Equation (4) shows that the extinction probability scales continuously with antibiotic concentration. To illustrate, consider two antibiotics, Azithromycin ($\alpha = 9.1$, $\beta = 1.12$) and Ciprofloxacin ($\alpha = 71.8$, $\beta = 2.46$). Equation (3) gives their critical concentrations: $c^*_{\text{Azi}} \approx 0.139$ μg/mL and $c^*_{\text{Cip}} \approx 0.175$ μg/mL. Azithromycin therefore guarantees eventual extinction at a lower dose.

While Theorem 2.6 guarantees eventual extinction whenever $\mu < 1$, it says nothing about how quickly the colony collapses. To compare the relative efficacy of two antibiotics operating in the subcritical regime, we examine the expected population at generation n. Because the total population at generation $t+1$ is a sum of i.i.d. copies of C, one per individual alive at t, the tower property gives $\mathbb{E}[B(t+1) \mid B(t)] = B(t)\,\mu$. Iterating from $B(0) = N_0$ yields

$$\mathbb{E}[B(n)] \;=\; N_0\,\mu^n \;=\; N_0\left(\frac{2}{1+\alpha\,c^{\beta}}\right)^n. \tag{6}$$

A smaller μ produces faster geometric decay, so $\mathbb{E}[B(n)]$ serves as a measure of comparative eradication speed between drugs. At $c = 0.18$ μg/mL, $\mu_{\text{Azi}} \approx 0.857$, and $\mu_{\text{Cip}} \approx 0.972$.

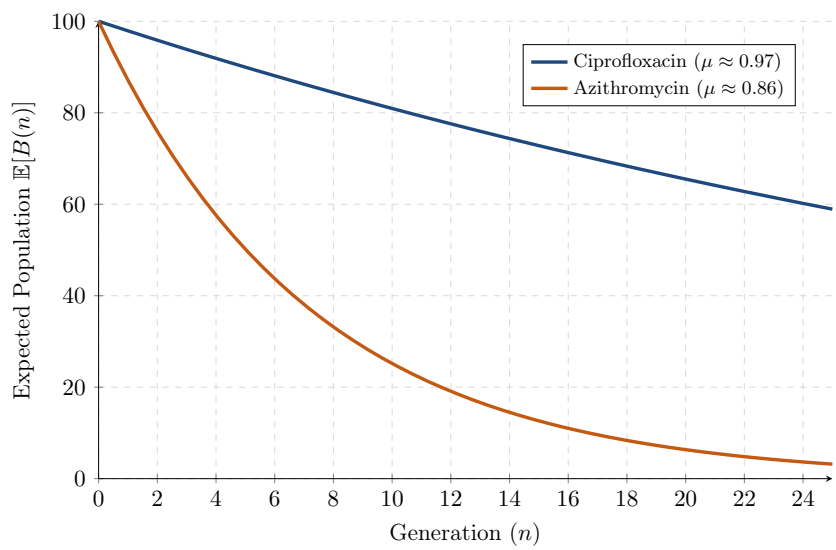

Figure 4: Theoretical mean colony size $\mathbb{E}[B(n)] = N_0\mu^n$ for $N_0 = 100$ cells at $c = 0.18$ μg/mL, with p computed from Assumption 3.1.

It is important to note that the generations are abstract, synchronous steps. The model tells us that Azithromycin drives the colony down faster per generation, but it cannot tell us how many minutes or hours a generation represents. Translating this comparative result into real-time eradication bounds would require a continuous-time formulation, as discussed in Section 5.2.

4 Simulation and Visualization

To model observable phenomena, we use Python and the Manim animation library [The Manim Community Developers] to simulate the branching process. These simulations generate branching trees based on the Bernoulli $\{0, 2\}$ offspring distribution. By running an ensemble of these stochastic processes simultaneously, we can empirically observe the colony sizes converging toward the theoretical expectations derived in Section 3. This approach allows for the dynamic visualization of eradication speeds (as seen in Figure 4) and provides an intuitive confirmation of the underlying theorems.

4.1 Encoding the Model

We begin by writing a function to compute the per-cell division probability, p, based on the function derived in Section 3:

$$p = \frac{1}{1 + \alpha\, c^{\beta}}\,. \tag{7}$$

This is encoded directly as a pure function in Python:

```python
def get_p_divide(c, alpha, beta):
    return 1.0 / (1.0 + alpha * (c ** beta))
```

Next, we construct a data dictionary to store the drug-specific parameters, α and β. We also assign a display color for the renderer, and define the treatment concentration $c > c^*$ alongside the initial population size N_0:

```python
DATA = {
    "Azithromycin":  {"alpha": 9.1,  "beta": 1.12, "color": BLUE},
    "Ciprofloxacin": {"alpha": 71.8, "beta": 2.46, "color": GREEN
        },
}
C_TREAT = 0.18         # treatment concentration (ug/mL)
START_POP = 100        # initial colony size N_0
```

Evaluating the `get_p_divide()` function at a treatment concentration of $c = 0.18$ μg/mL yields the per-generation division probabilities reported in section 3.4:

```
p_azi = get_p_divide(C_TREAT,
            DATA["Azithromycin"]["alpha"],
            DATA["Azithromycin"]["beta"])    # ~ 0.429

p_cip = get_p_divide(C_TREAT,
            DATA["Ciprofloxacin"]["alpha"],
            DATA["Ciprofloxacin"]["beta"])   # ~ 0.486
```

Because both p_{azi} and p_{cip} are less than $1/2$, both antibiotics result in $\mathbb{E}(C) = \mu = 2p < 1$ guaranteeing extinction by Theorem 2.6. However, the difference between the two expected offspring produces markedly different extinction trajectories.

4.2 The Branching-Process Engine

To perform the simulation, we write a function `get_next_gen()` that implements a single step of the branching process. The function draws a vector of independent Bernoulli trials for the entire colony simultaneously, determining for each cell whether it divides and produces a daughter cell with probability p, or fails to divide and is removed from the population with probability $1 - p$.

```
def get_next_gen(self, group, p, center, color, rng):
    if len(group) == 0:
        return [], [], []
    n = len(group.submobjects)
    survives = rng.random(n) < p
    parents = [c for c, s in zip(group, survives) if s]
    to_remove = [c for c, s in zip(group, survives) if not s]
    kids = []
    for c in parents:
        k = BacterialCell(c.get_center(), center, color=color)
        k.add_updater(lambda m, dt: m.drift(dt))
        kids.append(k)
    return to_remove, parents, kids
```

The function isolates the population into three disjoint lists: `to_remove` for cells that fail to divide, `parents` for the original cells undergoing fission, and `kids` for the newly created daughters. Structuring the output this way allows the animation layer to render births and deaths as distinct, sequential visual events within a single generational step.

4.3 Generational Loop and Population Tracking

We simulate the process over $G = 40$ discrete generations using a `for` loop. In each iteration, we apply the branching rule independently to the two antibiotic environments:

```
for g in range(1, 41):
    gen_tracker.set_value(g)

    rem_a, par_a, kid_a = self.get_next_gen(
        azi_cells, p_azi, dish_l_pos, BLUE, rng)
    rem_c, par_c, kid_c = self.get_next_gen(
        cip_cells, p_cip, dish_r_pos, GREEN, rng)

    # ... animation of splits and deaths ...

    azi_cells.add(*kid_a)
    cip_cells.add(*kid_c)
    azi_cells.remove(*rem_a)
    cip_cells.remove(*rem_c)

    if len(azi_cells) == 0 and len(cip_cells) == 0:
        break
```

At the end of each generation, the on-screen arrays are updated to reflect the new colony sizes, providing a real-time numerical readout. If both colonies reach extinction before the 40th generation the simulation terminates early.

4.4 Visual Design

The resulting visualization displays two side-by-side petri dishes, initialized with $N_0 = 100$ cells each. To simulate the natural movement of cells in a fluid environment, individual cells are modeled as point-particles undergoing a constrained drift with boundary reflection. For visual clarity, cell division is rendered by retaining the parent dot and spawning a single daughter rather than replacing the parent with two new dots. This is cosmetic; because the parent persists in place of one daughter, the population count each generation is identical to the formal $\{0, 2\}$ model. Cell death is represented by a shrinking fade-out. A full-screen flash marks the moment of antibiotic administration, after which the loop begins and the contrasting eradication efficiency of Azithromycin and Ciprofloxacin becomes apparent.
The complete source code is available at Umeadi [2026].

5 Conclusion

5.1 Summary

This paper explored the application of homogeneous branching processes to model bacterial proliferation via binary fission. By leveraging probability generating functions, we established a closed-form threshold for the critical antibiotic concentration c^* required to guarantee colony extinction. Furthermore, we demonstrated how this mathematical threshold translates to comparative clinical efficacy, showing that drugs with a lower c^* not only guarantee eradication at lower doses but also drive faster per-generation decay of the expected population.

5.2 Limitations and Future Directions

While the homogeneous branching process provides a robust foundational model, there are several simplifying assumptions: the antibiotic concentration is fixed, cells do not mutate or develop resistance, and there is no dormancy. Most critically, the model relies on discrete, generation-by-generation steps. In actual clinical scenarios, a physician is not just interested in the probability of a bacterium eventually becoming extinct after n abstract generations, but in the exact speed at which eradication occurs.

Future work should transition from this discrete-generation model to a continuous-time Markov branching process. By incorporating an exponential or Poisson-distributed lifespan for each individual cell, the model could establish specific time bounds for eradication. Additionally, future iterations of this model should account for biological complexities such as bacterial dormancy and the evolutionary development of antibiotic resistance over successive generations.

5.3 Acknowledgments

I thank my stochastic processes professor, Pierre Yves Gaudreau Lamarre, for his thorough and engaging introduction to the field, and for clearly demonstrating the practical utility of the course material. I also thank my probability theory and independent study professor, Justin Ko, for demonstrating the universality of probability theory. Finally, I thank the Manim Community Edition development team and community. The comprehensive documentation and responsive community support greatly facilitated the development of the simulation.

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